A. Buggy Sorting

time limit per test 2 seconds
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 900 | Tags: constructive algorithms, greedy, sortings | https://codeforces.com/problemset/problem/246/A

Little boy Valera studies an algorithm of sorting an integer array. After studying the theory, he went on to the practical tasks. As a result, he wrote a program that sorts an array of n integers $a_1, a_2, ..., a_n$ in the non-decreasing order. The pseudocode of the program, written by Valera, is given below. The input of the program gets number n and array a.

```
loop integer variable i from 1 to n - 1
    loop integer variable j from i to n - 1
        if (aj > aj + 1), then swap the values of elements aj and aj + 1
```

But Valera could have made a mistake, because he hasn't yet fully learned the sorting algorithm. If Valera made a mistake in his program, you need to give a counter-example that makes his program work improperly (that is, the example that makes the program sort the array not in the non-decreasing order). If such example for the given value of n doesn't exist, print -1.

Input

You've got a single integer n ($1 \le n \le 50$) — the size of the sorted array.

Output

Print n space-separated integers $a_1, a_2, ..., a_n$ ($1 \le a_i \le 100$) — the counter-example, for which Valera's algorithm won't work correctly. If the counter-example that meets the described conditions is impossible to give, print -1.

If there are several counter-examples, consisting of n numbers, you are allowed to print any of them.

Example

Input
1

Output
-1